

Computer Vision

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Question 1: Real-World Scene Capture to Digital Image Representation

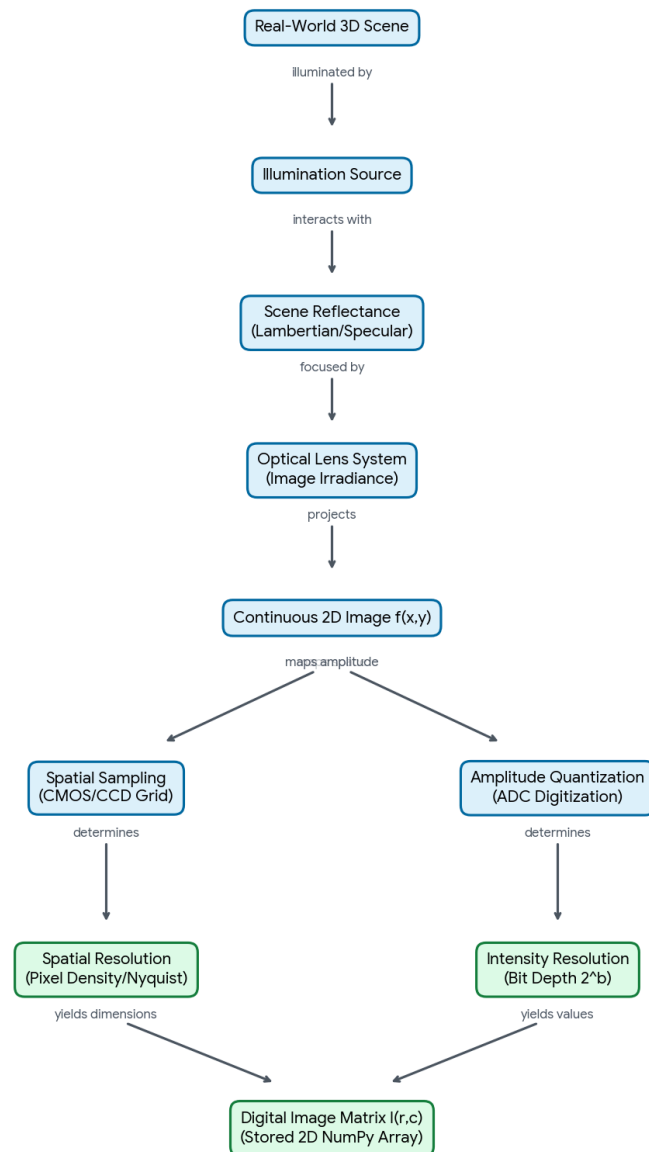
The physical-to-digital transformation pipeline bridges environmental physics and discrete computational matrix spaces. It operates through distinct steps:

1. **Scene Radiance and Reflected Light:** A real-world 3D scene is illuminated by an illumination source. The scene's surface reflects light based on its material properties, governed by Reflectance Models such as the Lambertian model (perfect diffuse reflection) or specular models.
2. **Optical Lens System:** The optical system collects these reflected rays and focuses them onto an imaging plane, converting scene radiance into image irradiance.
3. **Continuous 2D Image Formulation:** The projection creates a continuous 2D light distribution function, represented mathematically as $f(x, y) = i(x, y) * r(x, y)$, where i represents illumination and r represents reflectance.
4. **Spatial Sampling:** To digitize the continuous plane, a sensor grid (CMOS/CCD) samples the space at discrete intervals. This maps continuous spatial coordinates to pixel coordinates, establishing the Spatial Resolution.
5. **Amplitude Quantization:** The continuous electrical charge accumulated at each pixel is converted into a discrete digital value via an Analog-to-Digital Converter (ADC). This

maps the voltage spectrum into finite levels, establishing the Intensity Resolution (Bit Depth).

Concept Map Diagram:

Q1: Real-World 3D Scene to Digital Image Representation



Literature Evidence: Horn (1986), Szeliski (2022), Gonzalez & Woods (2018), Nakamura (2020), He et al. (2016).

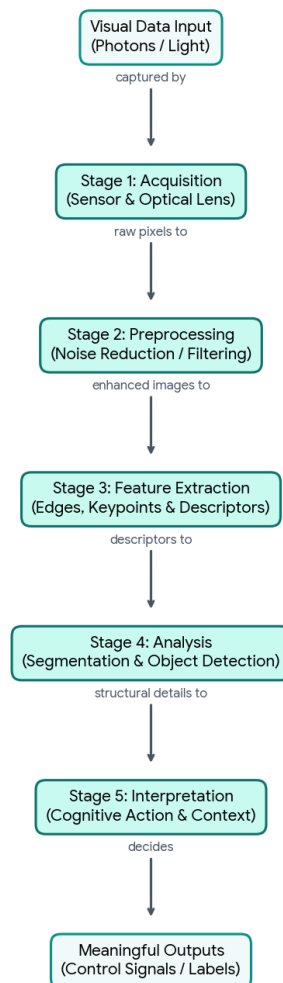
Question 2: Major Stages of Visual Data Handling in a CV System

An end-to-end computer vision pipeline transitions from raw physics to automated decision execution via five distinct abstraction layers:

- Stage 1: Acquisition: Hardware sensors transform light intensity into raw matrix data streams.
- Stage 2: Preprocessing: Prepares raw data by operations like noise reduction (Gaussian, Bilateral filtering) and histogram normalization.
- Stage 3: Feature Extraction: Low-to-mid level structural elements are isolated (e.g., edges, keypoints, SIFT, or CNN descriptors).
- Stage 4: Analysis: Isolated feature representations are grouped into regional patterns for segmentation or object detection (YOLO, Segment Anything).
- Stage 5: Interpretation: High-level semantic reasoning to execute final control signals or labels.

Concept Map Diagram:

Q2: Sequential Visual Data Handling Stages in CV Systems



Literature Evidence: Lowe (2004), Uijlings et al. (2013), Redmon et al. (2016), Badrinarayanan et al. (2017), Vaswani et al. (2017).

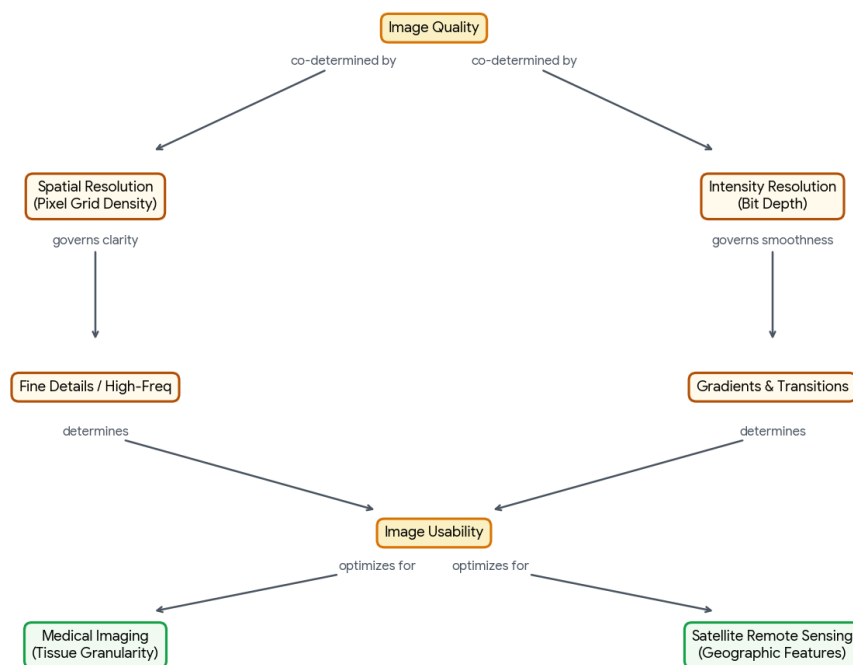
Question 3: Image Resolution vs. Image Quality Metrics

- Spatial Resolution: Defined by physical pixel density (DPI), determining the maximum spatial frequency captured and preserving fine structural elements.

- Intensity Resolution: Quantified by sensor bit depth, controlling the granularity of contrast levels and preventing visual false-contour banding.
- Image Quality: Metrics judge structural fidelity and artifact presence via Full-Reference (PSNR, SSIM) or No-Reference (NIQE, BRISQUE) methods.
- Domain Dynamics: Medical imaging prioritizes high intensity resolution (12-16 bit) to distinguish subtle tissue variation, whereas Satellite Remote Sensing prioritizes high spatial resolution for absolute geometric precision from altitude.

Concept Map Diagram:

Q3: Resolution Parameters & Image Quality Relationship



Literature Evidence: Shearer et al. (2019), Wang et al. (2004), Pratt (2014), Gevaert et al. (2015), Mittal et al. (2012).

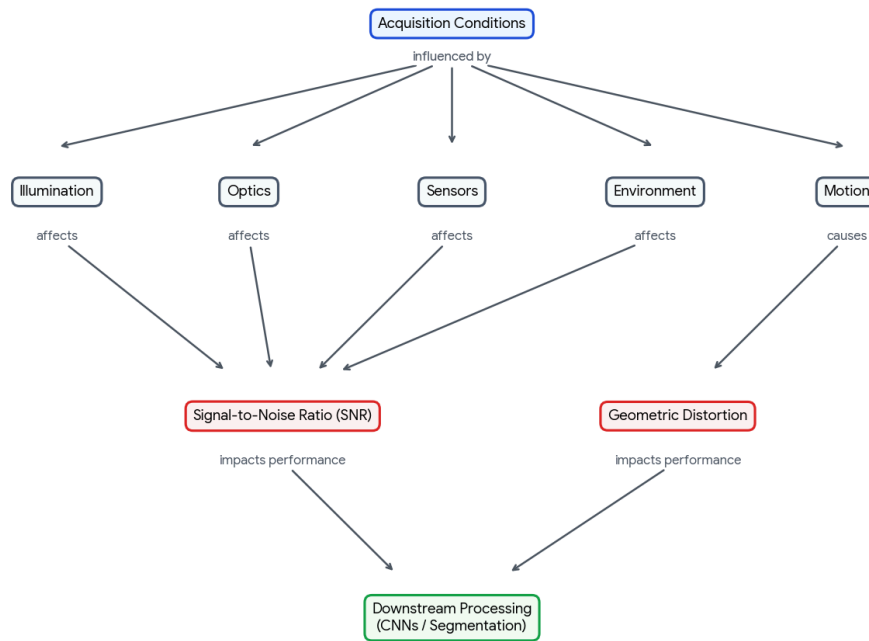
Question 4: Factors Influencing Image Acquisition Conditions

Real-world deployments introduce environmental variables that degrade performance prior to algorithmic computing:

- Illumination Fluctuations: Variations create shadows, specular blooming, or severe underexposure.
- Optics Degradations: Lenses introduce geometric aberrations (radial/tangential) and depth-of-field blur limitations.
- Sensor Limitations: Thermal activity introduces dark current noise and photon arrival variations (shot noise), degrading the SNR.
- Environmental Interferences: Atmospheric particulates (fog, rain, haze) scatter light, reducing contrast.
- Motion Instabilities: High relative velocity during exposure introduces motion blur kernels, acting as spatial low-pass filters.

Concept Map Diagram:

Q4: Factors Influencing Image Acquisition Conditions



Literature Evidence: Narasimhan & Nayar (2002), Schechner et al. (2001), Fergus et al. (2006), Reinhard et al. (2001), Cordts et al. (2016).

Question 5: CV Capabilities and Application Selection

Selecting a computer vision pipeline requires mapping a target domain's physical requirements to specific visual capabilities:

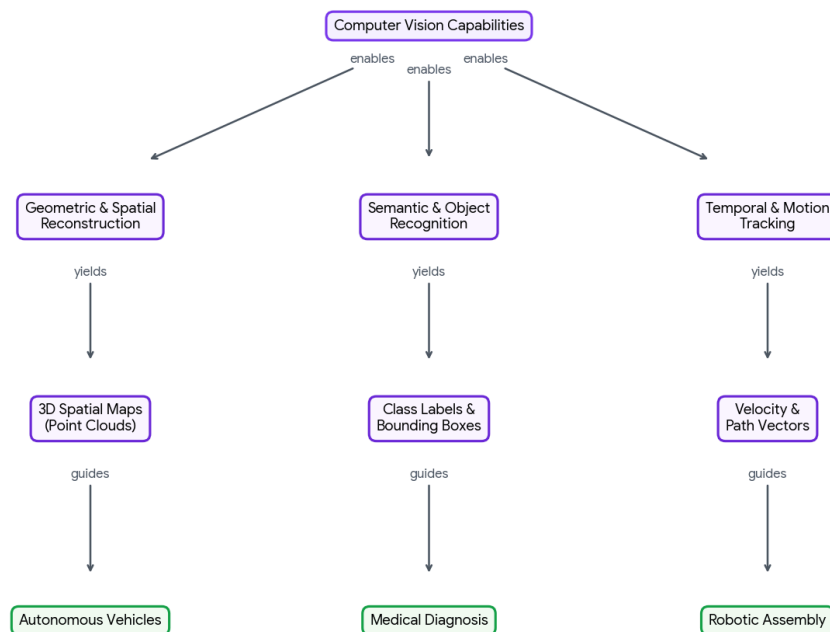
1. Geometric & Spatial Reconstruction: Focuses on tracking spatial coordinates via SfM or SLAM to extract dense 3D point clouds, essential for Autonomous Vehicles.

2. Semantic & Object Recognition: Parses categories and semantic structures via deep networks to output bounding boxes or masks, crucial for Medical Diagnosis.

3. Temporal & Motion Tracking: Extracts frame-to-frame optical flow fields or velocity profiles using Kalman/SORT tracking filters, vital for Robotic Assembly.

Concept Map Diagram:

Q5: System Capabilities vs. Application Selection



Literature Evidence: Cadena et al. (2016), Esteva et al. (2017), Bewley et al. (2016), Geiger et al. (2013), Krizhevsky et al. (2012).